



TED UNIVERSITY

CMPE 313/SENG 214

Software Engineering

Prestige Hotel Reservation

Software Requirements Specification (SRS) Document

SECTION 2 - TEAM 3

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1. Introduction

1.1 Purpose

This SRS document covers requirements of the Prestige Hotel Reservation system. This system includes login for 3 type of user (system admin, hotel admin and customer), hotel search , booking , hotel management for hotel owners and admins, and rating feedback for hotels.

1.2 Document Conventions

Standard IEEE 830 format is followed for requirement specifications. There are no special typographical conventions.

1.3 Product Scope

The system provides hotel listing, booking functionality, user ratings, filtering, and hotel management tools. This system provides us with easier and more secure access to finding available rooms in the hotels we are looking for. At the same time, it offers hotel owners a secure marketplace to promote their hotels.

1.4 References

Ref 1. Project Proposal Document (Prestige Hotel Reservation)

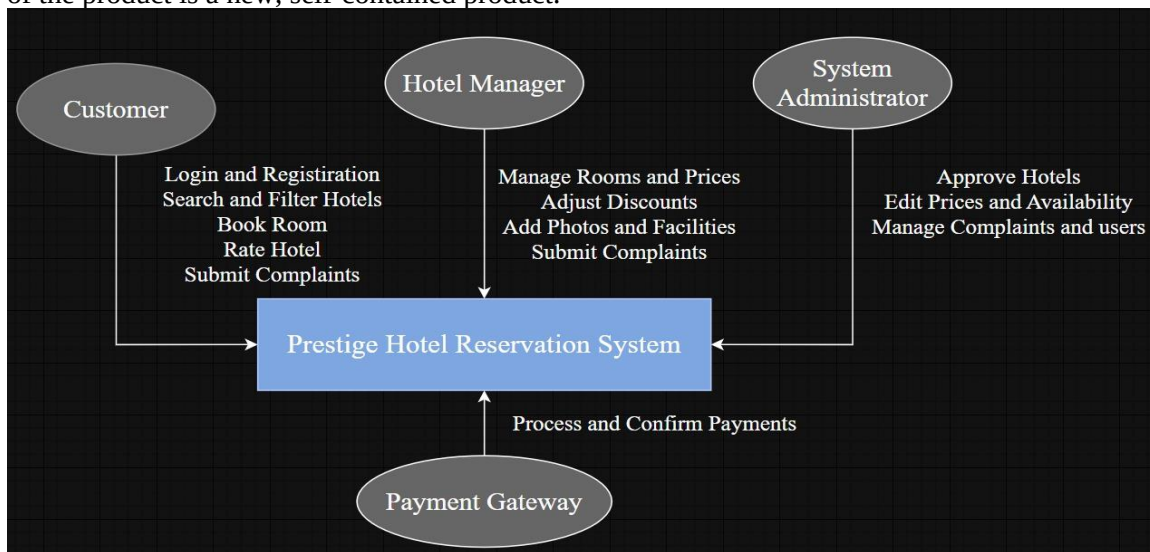
Ref 2. OpenAI. ChatGPT (2025). Prompt used: “How to write a proper srs document about hotel reservation system “.

Ref 3. IEEE 830 SRS Standard . Link: “<https://press.rebus.community/requirementsengineering/back-matter/appendix-d-ieee-830-sample/>”

2. Overall Description

2.1 Product Perspective

The system is a web-based application that integrates hotel search, booking, and management in a single platform. origin of the product is a new, self-contained product.



2.2 Product Functions

- Login and registration
- Hotel search based on hotel name
- Hotel search based on place, date, number of guests
- Result filtering based on price (min to max or max to min)
- Room booking functionality
- User hotel ratings functionality
- Hotel owner management tools such as adjusting room occupancy, price adjustments, discount adjustments, adding hotel photos, adding hotel facilities
- Administrative tools for hotel approvals and management

2.3 User Classes and Characteristics

- **Hotel Managers:** Manage room availability, prices, booking requests, reporting a complaint, request help, hotel request application (hotel declaration form),
- **Customers:** Search, filter, book rooms, provide ratings, reporting a complaint, request help, hotel, hotel request application (hotel declaration form),
- **System Administrator:** Approves hotels, can edit hotel price, room availability and manages user complaints, manage user roles (delete and authorize).

2.4 Operating Environment

OE-1: Web-based interface.

OE-2: Supports chromium-based browsers for both desktop and mobile.

2.5 Design and Implementation Constraints

DC-1: Developers are limited to utilize Microsoft's technology stack and must write the complete system in C# using the ASP.NET MVC framework (ideally .NET 6 or later).

DC-2: Schema designs must adhere to ACID standards for transactional integrity when using Microsoft SQL Server (MSSQL) as the relational database engine.

DC-3: Real-time data retrieval will be supported by Redis' implementation as an in-memory caching solution, which calls for the separation of durable database storage from volatile cache data.

DC-4: Because Auth0 enforces OAuth 2.0 and OpenID Connect protocols, all user authentication and authorization must be handled through it. This restricts the ability to create bespoke auth systems.

2.6 Assumptions and Dependencies

AS-1: We assume that the user is familiar with the use of common PC functions, including the mouse, keyboard, and standard Open and Save dialogs

AS-2: We assume that the user is familiar with the use of common mobile devices functions.

AS-3: We assume that the user is connected to a reliable internet connection.

AS-4: The system assumes that third-party services such as Google Maps and payment gateways will be operational and accessible via secure APIs.

DE-1: Third party services (Google Maps) will be available and functional.

DE-2: To manage user login, registration, token generation, and role-based access control, the system is dependent on the Auth0 authentication service's availability and uptime.

DE-3: The application must be able to contact the payment gateway provider (Iyzico) and facilitate safe, real-time transactions.

DE-4: For optimum performance, the system requires a Microsoft SQL Server environment that is reliable and configured with appropriate access permissions, backup plans, and indexing.

DE-5: To retain session data, enhance speed, and lessen database load, it depends on the Redis caching server operating as intended.

3. External Interface Requirements

3.1 User Interfaces

The system provides a very simple and straightforward web-based graphical user interface (GUI) designed for both customers and hotel managers. The interface includes:

Login Page: Users can sign up and login securely with registered email and password.

Registration page: The user can register with an e-mail address that has not been used before and the password that the user has determined, which he/she verifies with a verification code. It is mandatory to enter a phone number when registering and the user accepts the necessary KVKK document.

Home Page for Customers: Displays featured hotels, help button, manage reservation button, 2 search fields one for name-based and the other one for date, destination-based. The application requests cookies to remember user preferences, past searches, and login sessions to provide a more personalized experience.

Search and Filter Panel: It allows users to search by location, date and number of people, and to sort hotels by availability and customer ratings.

Home Page for Hotel Managers: upload hotel images, update room availability panel, adjust prices and apply discounts in a panel, and manage reservations panel and view.

Home Page for Administrators: Approve hotels panel and view, manage user roles panel, and handle complaints panel and view.

Hotel Details Page: Upon selecting a hotel, users are redirected to a detailed hotel page displaying; Room types and availability, pricing breakdown, guest reviews and ratings, hotel amenities and policies, Book Now button leading to the reservation process

Review & Rating Section: Customers can provide feedback and rate their hotel stays.

Booking Page: Customers can select their desired room type, input guest details, and proceed with a secure payment method. The system will confirm availability before finalizing the reservation. Users receive a confirmation email or SMS after a successful booking.

Help & Support Page: Users can submit complaints, request assistance, and view FAQs (frequently asked questions).

3.2 Hardware Interfaces

The Prestige Hotel Reservation system interacts with various hardware components such as Client-Side Devices and Server-Side Hardware. For the Client-Side Devices, the system is designed to function on desktops, laptops, tablets, and smartphones with internet access. However, for Server-Side Hardware, a dedicated or cloud-based web server is needed to host the application. Additionally, a database server for storing hotel, booking and user information is essential. Also, load balancers are needed for handling high traffic and ensuring smooth performance.

3.3 Software Interfaces

The Prestige Hotel Reservation system implements several software tools, solutions and libraries.

The system is intended to be accessed using any Chromium based browser (version 83.0.4103.97 or later) running on Windows 10 & 11 on PC, and Android 8.0 (or later) and Ios 11 (or later) on mobile.

The system also relies on Microsoft SQL Server Relational Database Management System (RDBMS) to implement its relational database to store and handle the storage and handling of information.

The system utilizes Google Maps to provide its users registered hotels' location data.

Login and authorization operations are handled through OAuth2.0 and OpenID Connect (OIDC) authorization and authentication frameworks.

All payments are processed through Iyzipay API.

Automated email confirmations and reminders are handled through Amazon Simple Email Service.

3.4 Communications Interfaces

All client-server interactions in the web-based Prestige Hotel Reservation system are based on common internet communication protocols. Among the communication interfaces are:

HTTP/HTTPS Protocols: These protocols will be used for all communications between the client devices and the server. For safe communication, HTTPS will be used, guaranteeing data integrity and encryption during the transfer process.

RESTful APIs: RESTful APIs over HTTPS will be used for communication between internal and external system components, such as payment gateways, authentication services like Auth0, and mapping services like Google Maps. These APIs guarantee uniform data sharing and follow the JSON language for message formatting.

WebSockets (Optional for real-time updates): WebSocket communication can be used to keep an open connection between the client and server for real-time functionality like live room availability updates or admin alerts.

Email Communication: To confirm registration, confirm reservations, change passwords, and provide other alerts, the system automatically sends out emails. This will be accomplished through the use of a secure and dependable third-party email provider (such as SendGrid or Outlook SMTP) using the Simple Mail Transfer Protocol (SMTP).

Security Protocols:

TLS encryption is required for any communications with external APIs (such as payment gateways and Auth0). For safe access control, OAuth 2.0 and JWT authentication tokens will be utilized in API headers. In order to stop DDoS assaults and API abuse, rate limitation and request validation will be implemented.

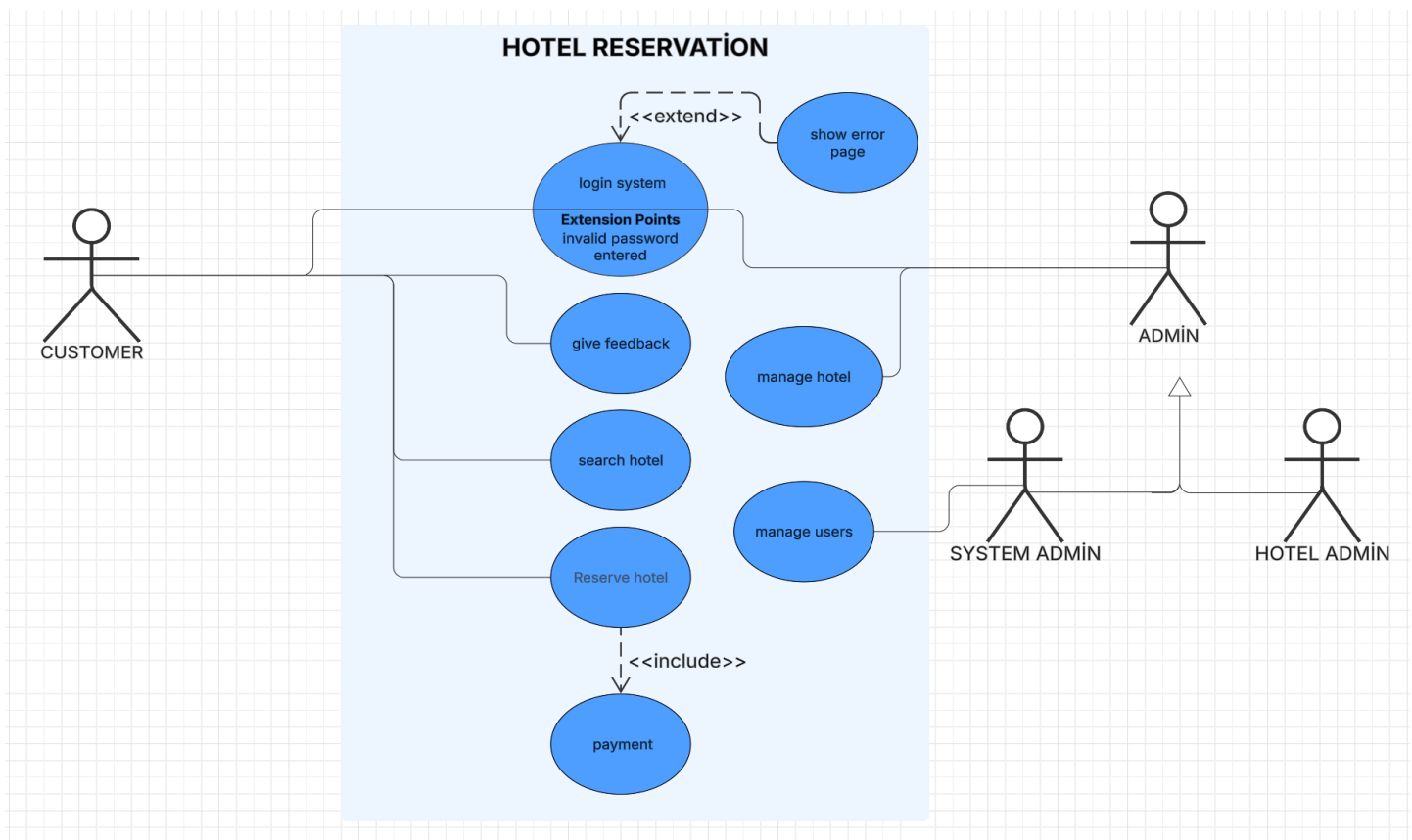
Synchronization of Data:

Redis will be utilized to reduce client request latency and cache frequently requested data.

Message Format: For readability, cross-platform compatibility, and processing simplicity, all data transferred over the system's RESTful services will be in UTF-8 encoded JSON format.

4. System Features

High Level Use Case Diagram



4.1 Hotel Reservation use case tables;

ID and Name:	UC-1 Login System
Actors:	Customer and Admin Secondary Actors: Authentication System
Description:	A user can access the system by logging in using their registered email and password. If the credentials are correct, access is granted. If the password is invalid, the system shows an error page.
Trigger:	A user wants to access his/her account for booking hotel
Preconditions:	PRE-1. The system is running actively and accessible for every user.

	PRE-2. The user has an active account which is registered and verified. PRE-3. The user has a stable internet connection
Postconditions:	POST-1. The registered card and personal information of the logged in user are updated and retrieved from the database. POST-2. The user is granted access to the "reserve hotel " POST-3. Login status is recorded for security
Normal Flow:	The user navigates to the login page of the system. They enter their registered email and password. The system validates the credentials via the authentication service. If the credentials are valid, the user is redirected to their home page.. The system displays a welcome message and provides access to reservation. If the user enters an invalid password, the system displays an error page informing them that their login failed.
Alternative Flows:	If the user forgets their password, they can initiate a password reset with email. If the user does not have an active account, the user can register and create an account with their personal information.
Exceptions:	If the login password and email are incorrect, the system blocks access with an error message.
Non-functional requirements	Requests for logins should be processed by the system in two seconds. At least 500 concurrent login requests should be supported by the system. Encrypting password entries and adhering to secure storage standards are essential. For audit and security review purposes, every login attempt should be recorded.

Functional Requirements UC1:

REQ-1: The system shall authenticate users using their registered email and password.

REQ-2: The system shall display an error message if login fails due to incorrect credentials.

REQ-3: The system shall allow users to reset their password via email if forgotten.

REQ-4: The system shall offer account registration for unregistered users.

REQ-5: The system shall allow access only if credentials are valid.

REQ-6: The system shall redirect valid users to their homepage.

REQ-7: The system shall display displays a welcome message if the login is completed successfully

ID and Name:	UC-2 Manage Hotel
Actors:	Admin
Description:	Every admin of the system can manage any hotel, can view the hotel that the admins have permission for. They can add a new hotel, can remove the hotel that is under their permission and change the features of the hotel that they have permission for.
Trigger:	The actor presses the manage the hotel button. If the actor wants to add hotel, actors press the add hotel button, if the actor wants to remove or edit the hotel, presses edit button
Preconditions:	PRE-1. Actor is logged into system. PRE-2. The actor must be using The Admin account. PRE-3. The actor has to have the permission for managing the specific hotels,
Postconditions:	POST-1. Initial hotel is added to the systems database under the hotel manager who adds the hotel and under the system admins with higher permissions. POST-2. The wanted changes are handled by the system for systems database. POST-3. If some hotels are deleted, the deleted hotels are removed from the systems database.
Normal Flow:	The user enters the main page of their admin page and presses the manage hotel button. They may want to add a new hotel in order to add, they press the add new hotel and after then admin adds the hotel information related to systems and customers' needs. Systems adds the

	hotel by the given information with admins requests. However, they may want to remove or change information for a specific Hotel.
Exceptions:	If the admin attempts to add a hotel with the same name of an existing hotel, the system gives an error message and does not change anything.
Non-functional requirements	The system should list the hotels that is under the hotel admin in ten seconds. Admin should receive a confirmation email after successful cancellation. Admins should receive a confirmation email after adding a new hotel. Admins should receive a confirmation email after changing feature of an existing hotel. Admins should receive a confirmation email after removing an existing hotel. Data updates should reflect in the system in real-time.

Functional Requirements UC2:

REQ-1: System shall allow hotel admins to add new hotel under their names.

REQ-2: System shall list of existing hotels under the names of hotel admin.

REQ-3: System shall allow system admins to add new hotel under their any hotel admins name.

REQ-4: Hotel managers and System admins must be able to update hotel and room details.

REQ-5: The system shall give error page if there is an existing hotel while admins tries to add new hotel with the same name.

REQ-6: Hotel admins shall be able to create promotional campaigns or early booking discounts for specific time periods. These campaigns shall be saved in the system with start and end dates.

REQ-7: The system shall allow hotel admins to update room availability, status (e.g., maintenance), and inventory in real time.

REQ-8: The system shall allow hotel admins to set and update room pricing per night, per room type, and based on season or campaigns.

ID and Name:	UC- 3 Manage Users
Actors:	System Administrator; Secondary Actors: Authentication System
Description:	The system administrator can add, edit, and delete user accounts, reset passwords, and manage user roles and permissions within the system.
Trigger:	Administrator accesses the user management section of the admin panel.
Preconditions:	PRE-1. Administrator is logged into the system. PRE-2. Administrator has appropriate system privileges to manage users. PRE-3. The authentication system is functioning properly.
Postconditions:	POST-1. User account details are updated in the database. POST-2. Notification is sent to users regarding account changes when appropriate. POST-3. System logs record all user management actions for security audit purposes.
Normal Flow:	- Administrator navigates to the user management panel from the main admin screen. - The system displays a list of all users with filtering and search capabilities. - Administrator can select a user to view their complete profile details. - Administrator can modify user information (name, contact details, status). - Administrator can change user roles. - Administrator can disable accounts that violate system policies. - Administrator can reset a user's password which triggers an email to the user. - System confirms all changes before they are committed to the database.
Alternative Flows:	- Administrator can create a new user account by providing necessary information. - Administrator can export user data for reporting purposes. - Administrator can view user activity logs to track suspicious behavior. - Administrator can manage role-based permissions across the system.
Exceptions:	- If attempting to delete an account with active reservations, the system warns the administrator and requires confirmation. - If attempting to give elevated privileges to a user who doesn't meet requirements, the system displays an error.

	3. If the email service is unavailable when resetting a password, the system stores the action and retries later
Non-functional requirements	1. The system must maintain an audit log of all user management actions. 2. User data must be encrypted in the database according to data protection regulations. 3. The user management interface should be responsive with actions completed in under 2 seconds. 4. Password reset emails should be sent within 1 minute of the reset action.

Functional Requirements UC3:

REQ-1: System shall allow administrators to view a complete list of all registered users.

REQ-2: The system shall allow editing of user profiles, including contact and role information.

REQ-3: System shall allow administrators to reset user passwords, triggering an automated email notification to the user.

REQ-4: The system shall validate actions that change user privileges and display errors if conditions are not met.

ID and Name:	UC-4 Reserve Hotel
Actors:	Customer; Secondary Actors: Hotel Inventory System
Description:	The logged-in user reserves the desired room and day of the hotel he/she chooses and reserves the available room by paying the fee.
Trigger:	The customer initiates the hotel reservation process.
Preconditions:	PRE-1. Customer is logged into system. PRE-2. The user has an active account. PRE-3. The hotel that the customer wants to reserve has a room available. PRE-4. The customer must have sufficient money in his/her bank account at the purchase.
Postconditions:	POST-1. The hotel's vacant and occupied rooms are updated with the latest purchases or cancellations. POST-2. Purchase or cancellation notification message is sent to the hotel admin and the user simultaneously. POST-3. Reservation information and user information are recorded in the system for possible errors and security reasons.
Normal Flow:	The user who is logged into the system can examine the hotel he/she has chosen. Then he/she can make a reservation for the hotels that have availability. He/she is directed to the payment page for the reservation and after completing the payment correctly on this page, he/she receives an information message that this reservation is confirmed. There is a star system where a user who has used the hotel before can evaluate the hotel, the user can make this evaluation if he/she wants.
Alternative Flows:	If a preferred room is unavailable, users can be suggested alternative rooms or dates. If room availability changes during the process, customer is notified. Users can cancel their reservation within a refundable timeframe. Hotel managers can update room details, pricing, and availability.
Exceptions:	If the user attempts to book an already reserved, the system displays a real-time availability update. If the reservation is non-refundable, the system prevents cancellation and informs the user. If the payment fails, the reservation is canceled, and the user is notified. If user lost their internet connection in process ,reservation process stops and checks logs for safety reasons and and makes the necessary refund transactions.
Non-functional requirements	The system should support at least 100 concurrent bookings. Users should receive a confirmation email after successful cancellation. All payment transactions must be protected through encryption to ensure data security The system shall update room availability in real-time.

Functional Requirements UC4:

REQ-1: The system shall allow customers to view room availability for selected hotels.

REQ-2: The system shall display available room types and their pricing based on selected dates.

REQ-3: The system shall allow logged-in customers to reserve rooms for selected dates.

REQ-4: The system shall confirm successful bookings and display a confirmation message.
 REQ-5: The system shall notify both the customer and the hotel admin after a successful booking or cancellation.
 REQ-6: The system shall log reservation details for security and support purposes.
 REQ-7: The system shall validate that selected dates are in the future and the check-out date is after the check-in date.
 REQ-8: The system shall update room availability in real-time after a successful reservation.
 REQ-9: The system shall validate the user's bank account or credit card for sufficient funds.
 REQ-10: The system shall cancel the reservation automatically if payment is not completed.
 REQ-11: The system shall display the availability status (Available / Fully Booked) of each room type for the selected hotel and dates.
 REQ-12: The system shall retrieve and show the latest room pricing based on date, room type, and campaign status.
 REQ-13: During the reservation process, users shall automatically benefit from active campaigns and early booking discounts if applicable.

ID and Name:	UC-5 Search Hotel
Actors:	Customer; Secondary Actors: Hotel Inventory System
Description:	A customer can search for available hotels based on location, dates, price range, and other filtering criteria to find suitable accommodation options.
Trigger:	Customer initiates a hotel search using the search interface.
Preconditions:	PRE-1. Customer has access to the system (logged in or as a guest user). PRE-2. Hotel database is functional and updated with current information. PRE-3. Search functionality is operational.
Postconditions:	POST-1. System displays search results that match the customer's criteria. POST-2. Search parameters and results are temporarily stored for refinement. POST-3. Search activity is logged for analytics and personalization purposes.
Normal Flow:	1. Customer enters the search criteria (location, check-in/check-out dates, number of guests). 2. Customer submits the search request. 3. System queries the database for hotels matching the basic criteria. 4. System displays a list of available hotels with basic information (name, thumbnail image, star rating, base price). 5. Customer can view more details about any hotel in the search results. 6. Customer can select a hotel to proceed with the booking process.
Alternative Flows:	1. Customer can apply additional to narrow search results (price range, user ratings). 2. Customer can sort results by different criteria (price low to high, price high to low). 3. Customer can modify search parameters and resubmit the search. 4. Customer can save search criteria for future if logged in. 5. Customer can view hotels on a map if location-based search is used.
Exceptions:	1. If no hotels match the search criteria, the system displays a message and suggests modifying search parameters. 2. If the search criteria are invalid (e.g., check-out date before check-in date), the system displays an error message. 3. If the search functionality is temporarily unavailable, the system notifies the customer to try again later. 4. If too many results are returned, the system may paginate results or suggest narrowing the search.
Non-functional requirements	1. Search results should load within 3 seconds for typical queries. 2. The system should support at least 100 concurrent search operations. 3. Search functionality should be available 99.9% of the time. 4. The search interface should be responsive and work on both desktop and mobile devices. 5. Recent searches should be cached to improve performance for repeated queries.

Functional Requirements UC5:

REQ-1: The system shall allow users to input search criteria such as location, dates, and number of people.
 REQ-2: The system shall return a list of matching hotels based on the input criteria.

REQ-3: The system shall allow sorting the results from expensive to cheap or cheap to expensive.

REQ-4: System shall suggest alternative dates or nearby locations if no results match the original search criteria.

REQ-5: The system allows hotel search based on hotel name.

ID and Name:	UC-6 Give Feedback
Actors:	Customer
Description:	A customer can rate a hotel and gives feedback for reviewing
Trigger:	A customer gives feedback from a visited hotel and rate by hotels page
Preconditions:	PRE-1. Customer is logged into system. PRE-2. The user has an active account. PRE-3. The Hotel must be visited before by that specific customer.
Postconditions:	POST-1. New feedback and rate added the system. POST-2. Rate and Feedback as comment added the hotel page
Normal Flow:	They will choose from visited hotel page. After choosing, stars for rating and a textbox appears for review. After submitting the rating and feedback, hotel page will be updated
Alternative Flows:	If the added feedback consist the insult or bad language, comments will be seen by an system admin and will be removed.
Exceptions:	If the customer tries to the rate or give feedback that he/she did not visit, system will show a warning message then will prevent the event from happening.
Non-functional requirements	The system should update the hotel page in ten second. Users should receive a confirmation email after successfully submitting the feedback and rate. Data updates should reflect in the system in real-time.

Functional Requirements UC6:

REQ-1: The system shall display an interface in which there will be five stars for rating and text field for feedback.

REQ-2: The system shall update the specific hotel page according to new rating and feedback.

REQ-3: The system shall send feedback for moderation if it contains inappropriate content.

REQ-4: The system shall allow user to edit their comments and review if they want to edit.

REQ-5: The system shall prevent feedback submission for hotels not visited by the user.

5. Nonfunctional Requirements

5.1 Performance Requirements

- Database should be updated every 3 seconds.
- Query results should load within no longer than 4 seconds.
- User Interface should not take more than 2 seconds to load.
- Login validation should be complete within 3 seconds.
- Website should support at least 500 online users.
- Payment confirmation process should be complete in 5 seconds.

5.2 Safety Requirements

- All communications between data server and client must be encrypted.
- All data shall be stored and protected or protectively marked.
- Database should be backed up every hour.
- Under a fail situation, the system should be operational again within one hour.
- Administrators cannot access personal information like card id, security code and etc.

5.3 Security Requirements

- Payment system shall integrate 3D secure for transactions.
- If the user forgets her/his password, she/he can reset her/his password via e-mail address.
- Users should be able to connect their accounts to a secondary recovery email in case they lose access to their primary one.
- Passwords shall accept alphanumerical characters.
- Passwords shall not be less than 8 characters long.

5.4 Software Quality Attributes

- **Correctness:** The system must meet the end user objectives to fully perform normal regular Hotel Management operations.
- **Efficiency:** Applicate efficient resources to accomplices a particular task efficiently and without a hassle.
- **Availability:** Progress report should be generated automatically in every month for system administrators.
- **Usability:** System should have user manual for every actor.
- **Flexibility:** System should be flexible enough to provide space to add new features and handle them comfortably.
- **Portability:** The system must work on in every chrome-based search engine and Microsoft (Windows) environment.
- **Integrity:** The system should focus on securing the customer information and avoid data lose as much as possible
- **Interoperability:** The system should provide an environment where hotel managers and system administrators can work without interruption.
- **Maintainability:** The system should be open %99.9 of the month.
- **Robustness:** The System have to be protected itself from DDOS attacks.

6. Other Requirements

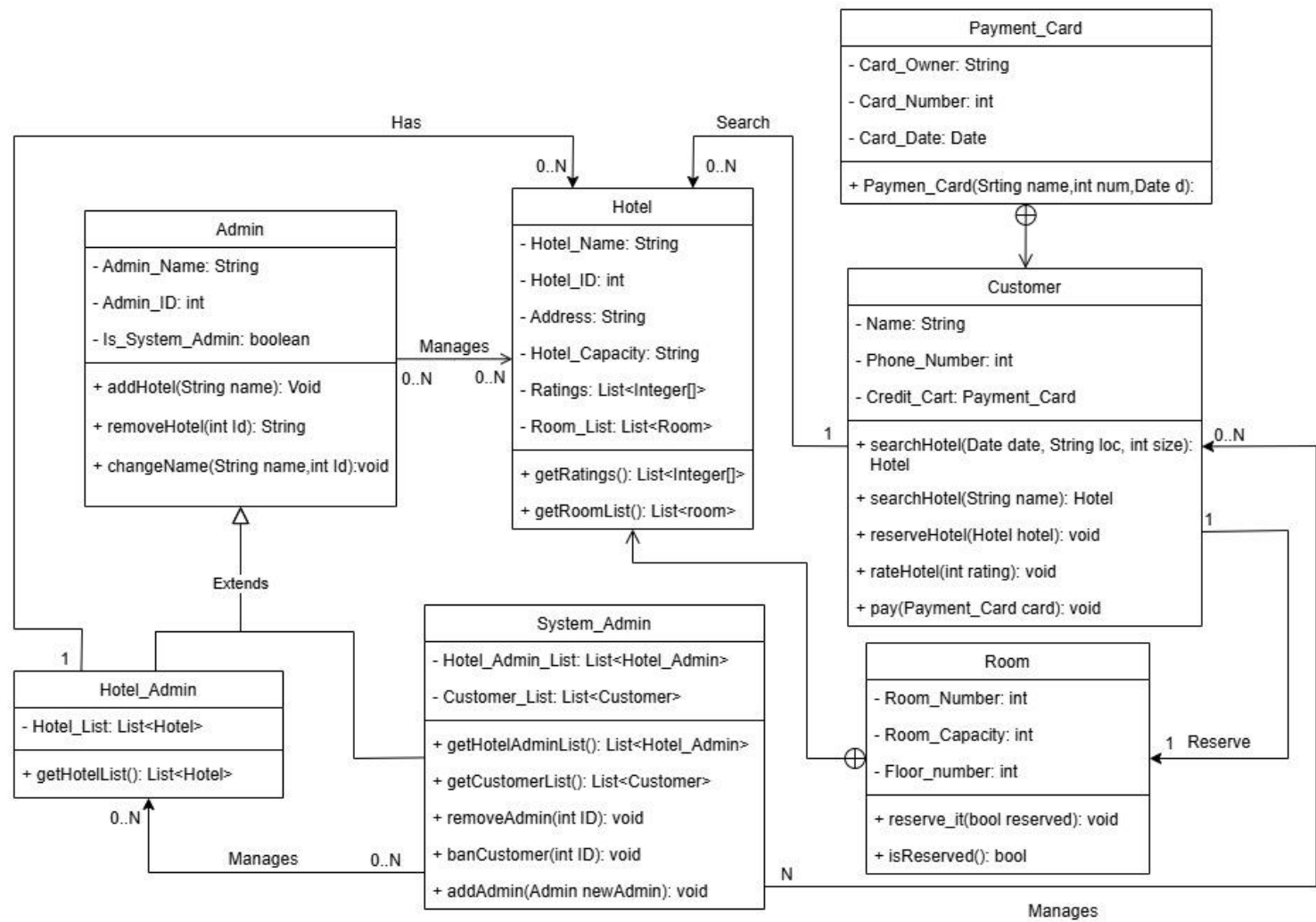
Appendix A: Glossary

SRS	Software Requirements Specification
RBAC	Role-Based Access Control - A security model for controlling user permissions
ACID	Atomicity, Consistency, Isolation, Durability - Principles for database integrity
API	Application Programming Interface; a set of protocols for building and integrating application software.
Auth0	A third-party authentication and authorization service used for secure login and user management.
Iyzico (Iyzipay)	The online payment gateway used to securely process customer payments.
JWT (JSON Web Token)	A compact token format used for securely transmitting information between parties as a JSON object.
OAuth 2.0	An open standard for access delegation, commonly used for token-based authentication.
KVKK	Personal Data Protection Law - Data privacy regulation in Turkey
OpenID Connect	A simple identity layer on top of OAuth 2.0, used for verifying user identities.
Redis	An in-memory data store used for caching and real-time data retrieval.
RESTful API	An API that conforms to the constraints of REST architecture, enabling interaction between systems over HTTP.
TLS (Transport Layer Security)	A cryptographic protocol designed to provide communications security over a computer network.

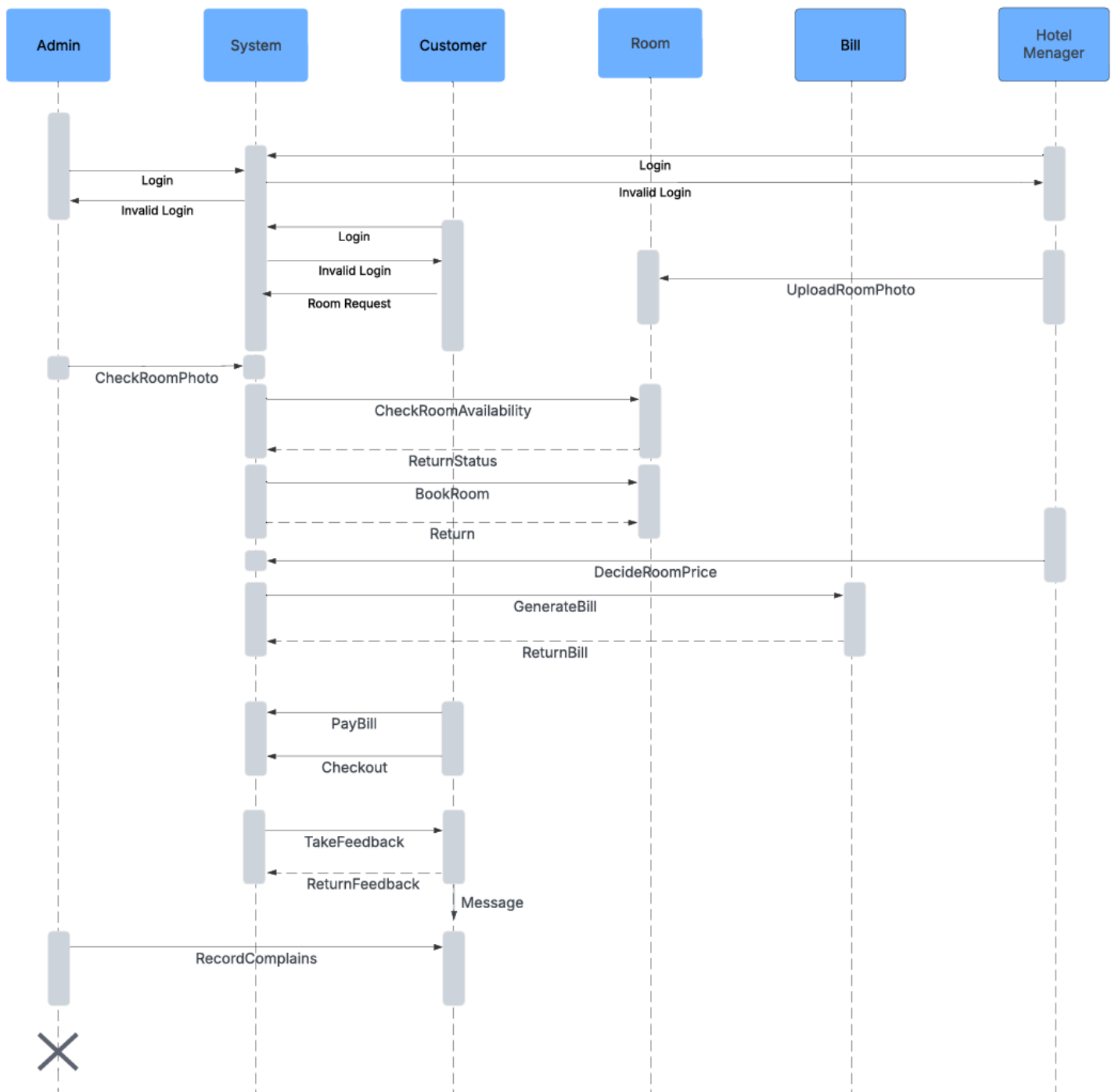
UI (User Interface)	The graphical layout and elements through which users interact with the system.
UX (User Experience)	The overall experience a user has when interacting with the system, including usability and satisfaction.

Appendix B: Analysis Models

Class Diagrams:



Sequence Diagrams:



Activity Diagram:

